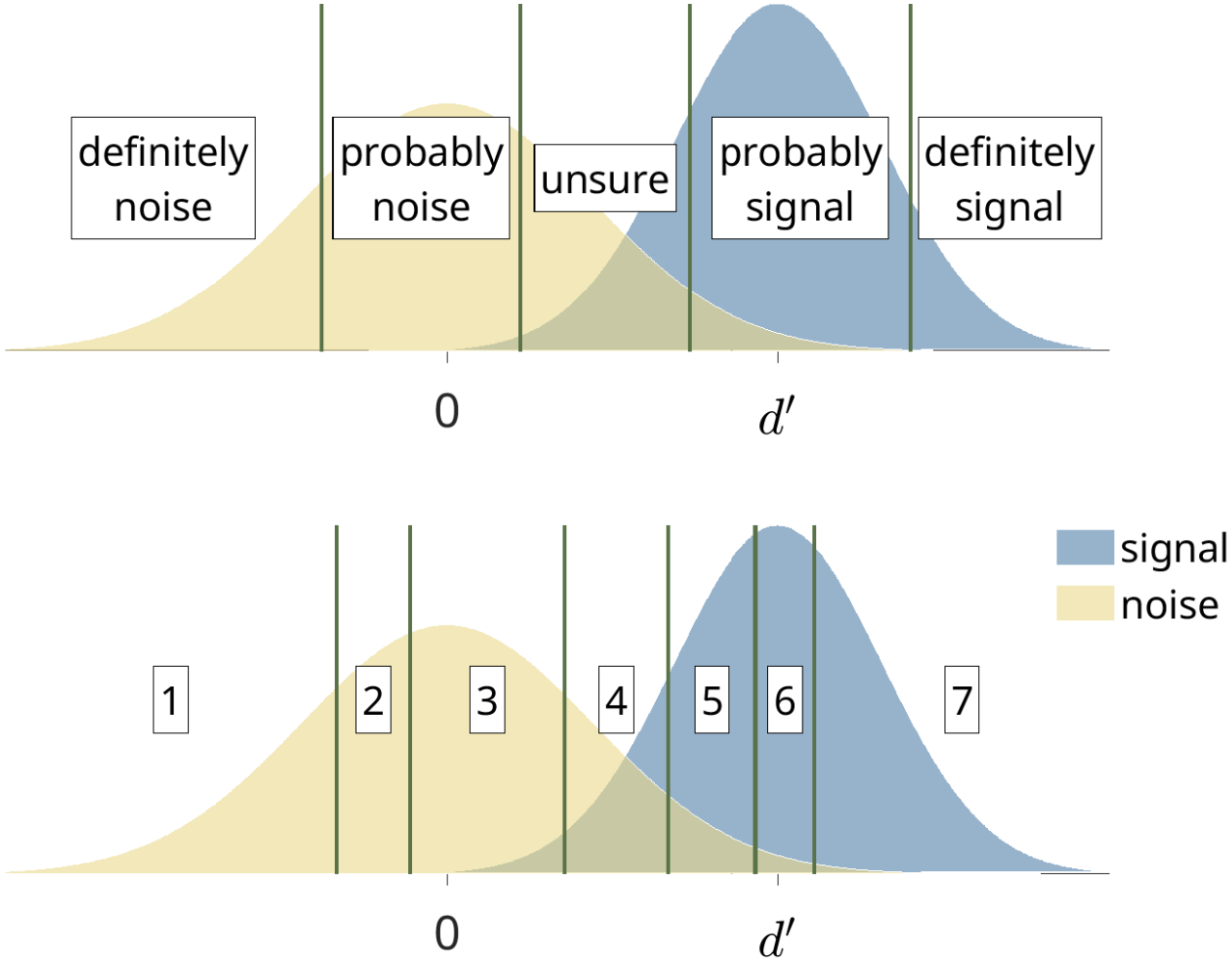
Multiple-Choice Signal Detection Model

Michael D. Lee

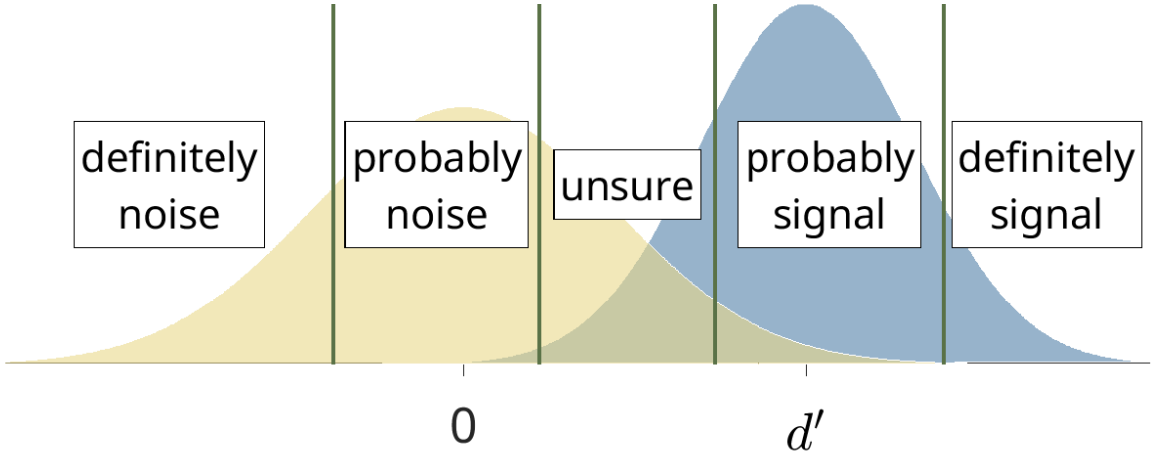
Multiple-Choice Signal Detection Model

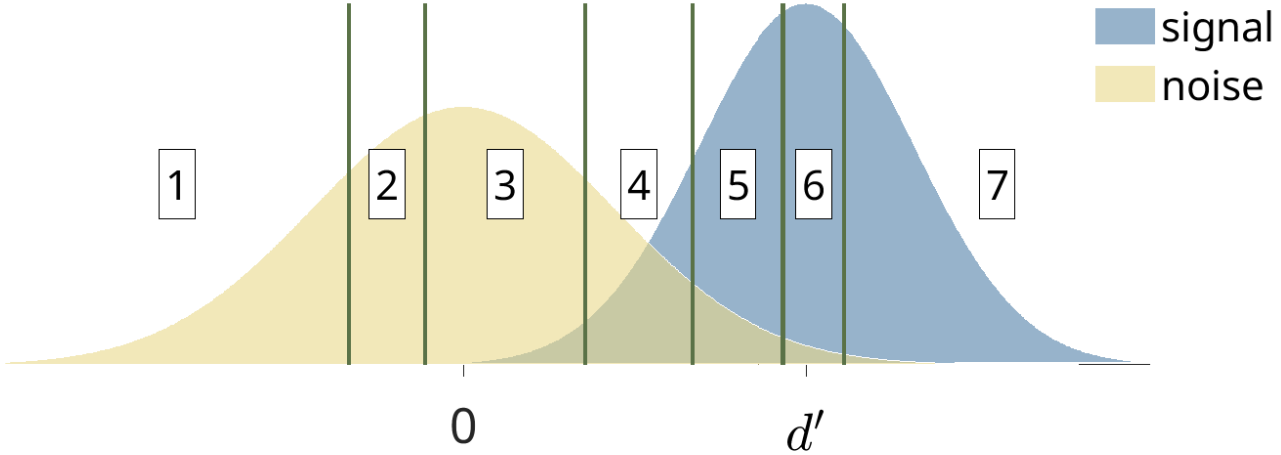
- The multiple-choice signal detection model allows for responses that express some level of confidence in detecting signal and noise



Multiple-Choice Signal Detection Model

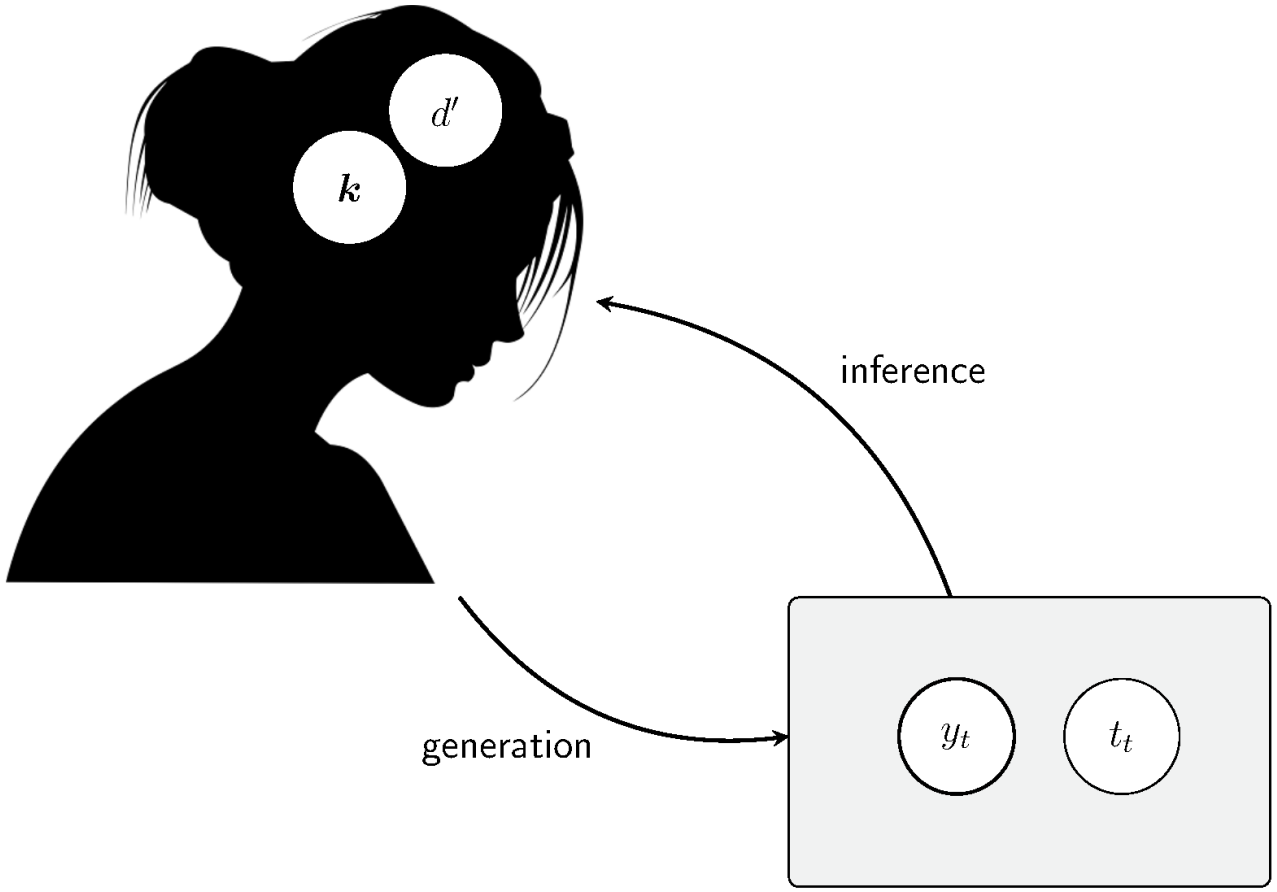
- Uses the same mental sampling process as the two choice model
  - now the mental sample from the signal or noise distribution falls into one of many response regions





Multiple-Choice Signal Detection Model Structure

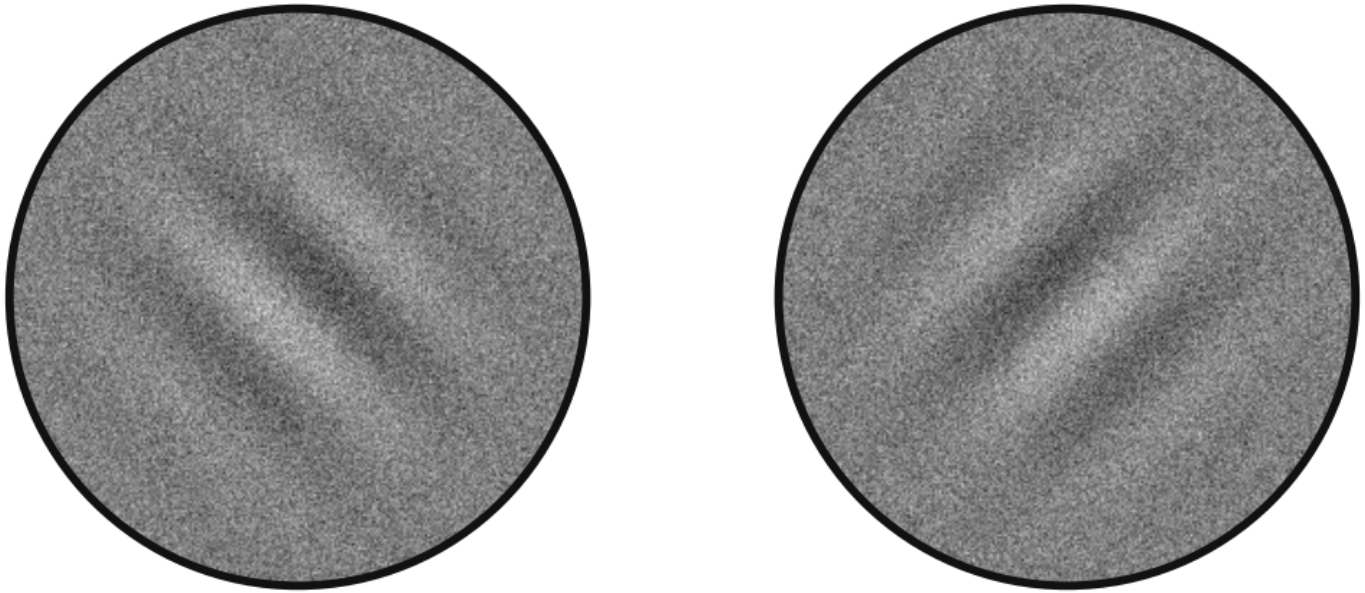
- The multiple-choice signal detection model has the structure
  - observed binary decisions y_t on trial t with the truth t_t indicating signal or noise
  - discriminability d' and a vector of criteria $\boldsymbol{k} = (k_1, k_2, \ldots)$ parameters



Visual Metacognition

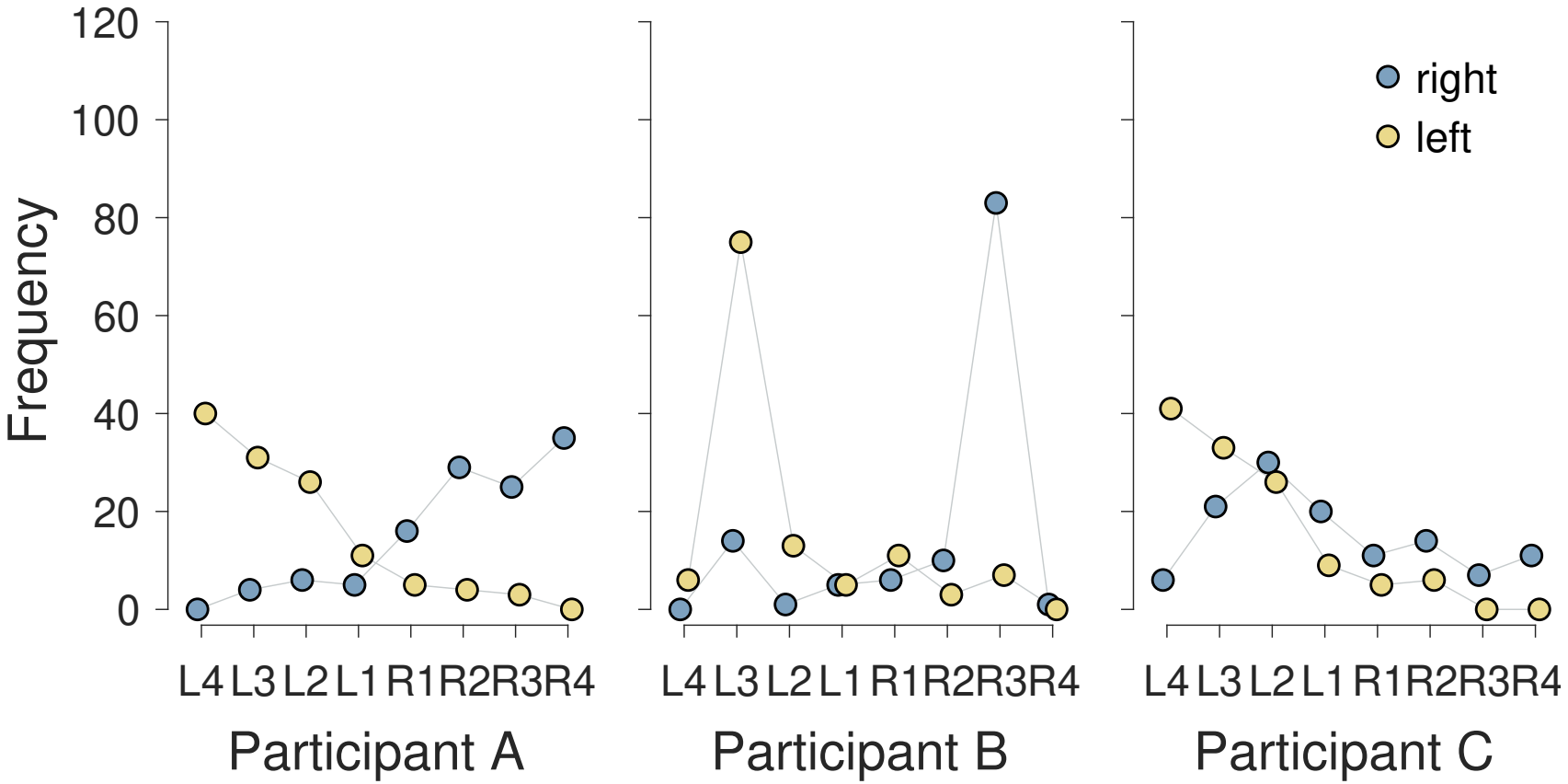
Visual Metacognition Experiment

- Shekhar and Rahnev (2018) conducted an experiment in which participants chose beween two possible orientations of Gabor stimuli
  - single response keypress indicating decision and confidence
  - hand used indicates decision
  - finger used indicates confidence, with index finger being the lowest confidence of 1 and little finger being the highest confidence of 4



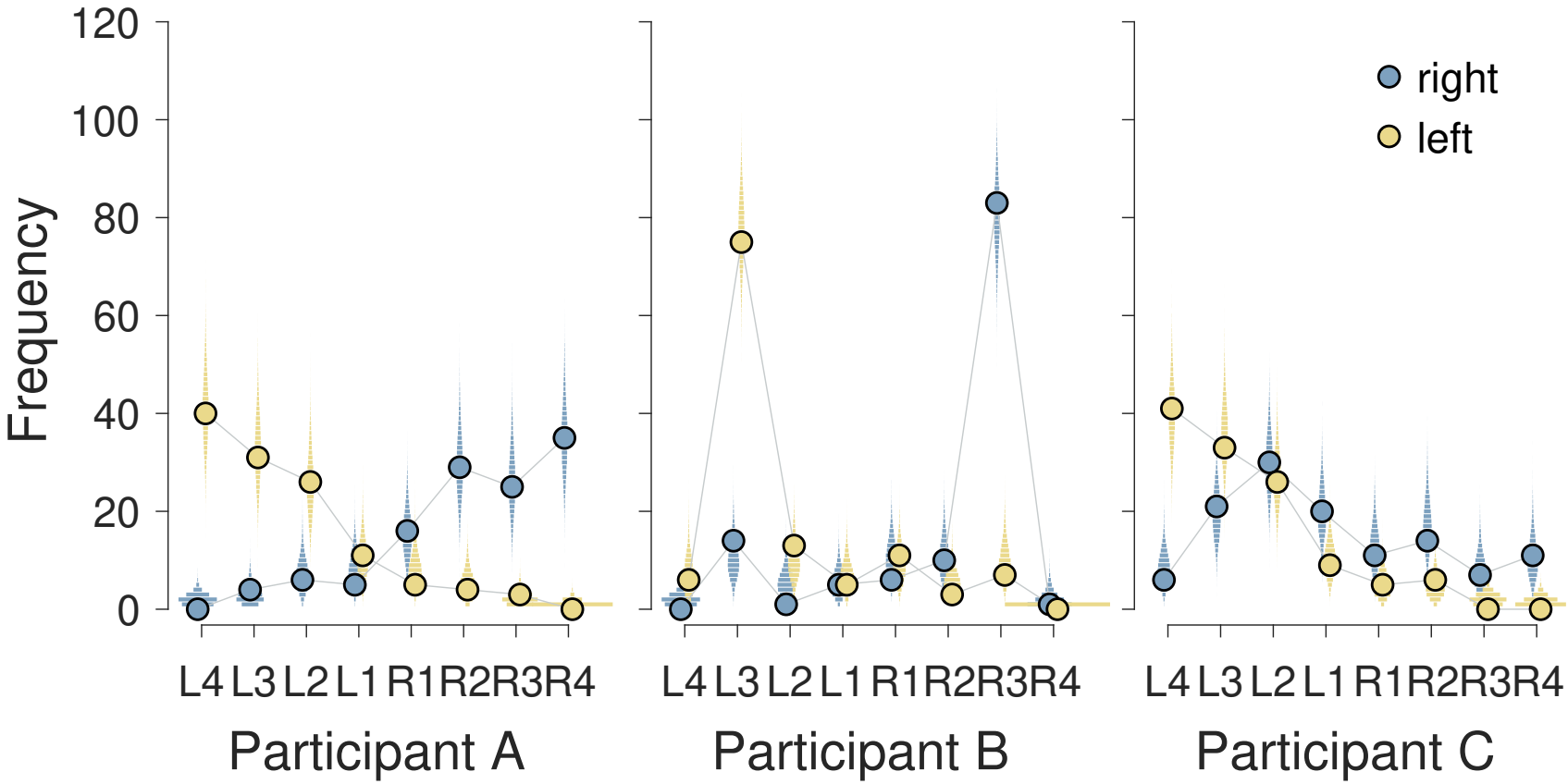
Data For Three Participants

- The figure summarizes the behavioral data from three participants
  - frequency of each of 8 possible responses
  - for both right-tilted and left-titled stimuli



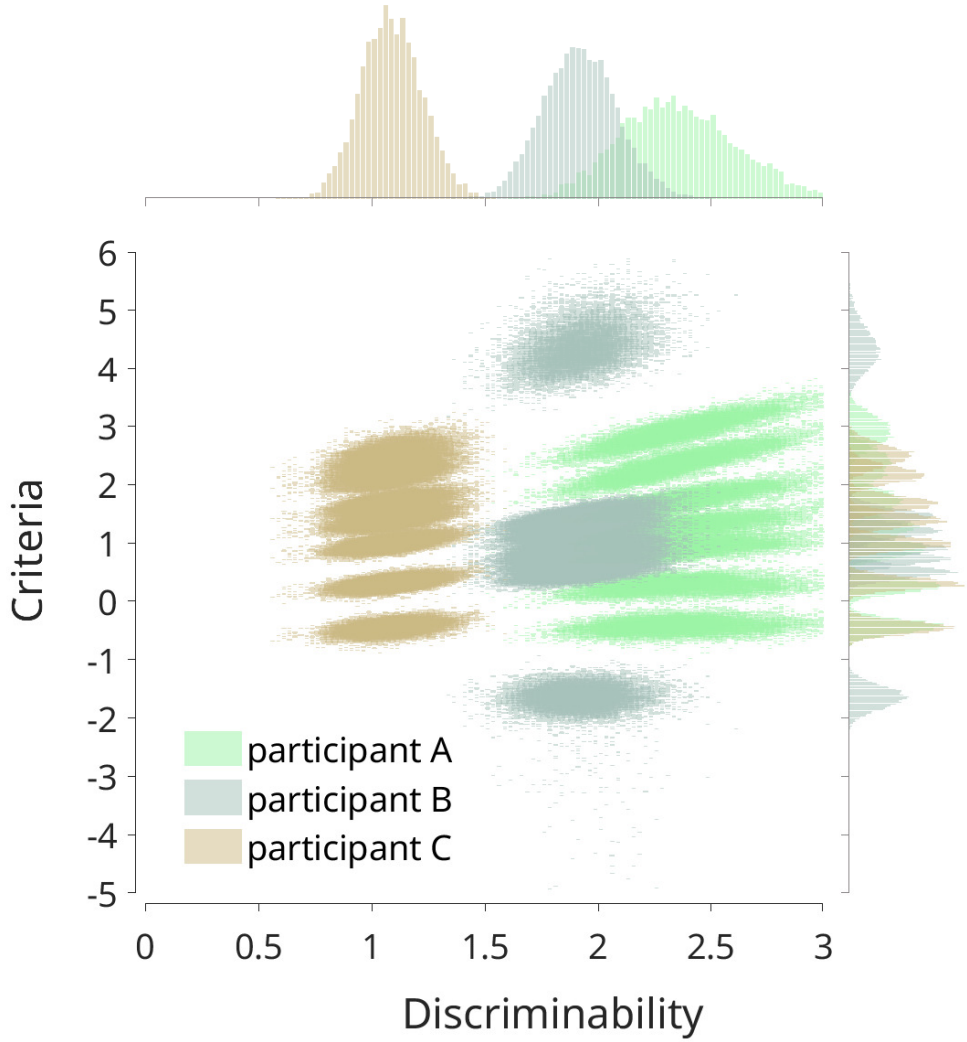
Descriptive Adequacy

- The posterior predictive distribution of the multiple-choice signal detection model based on the same experimental design shows good descriptive adequacy



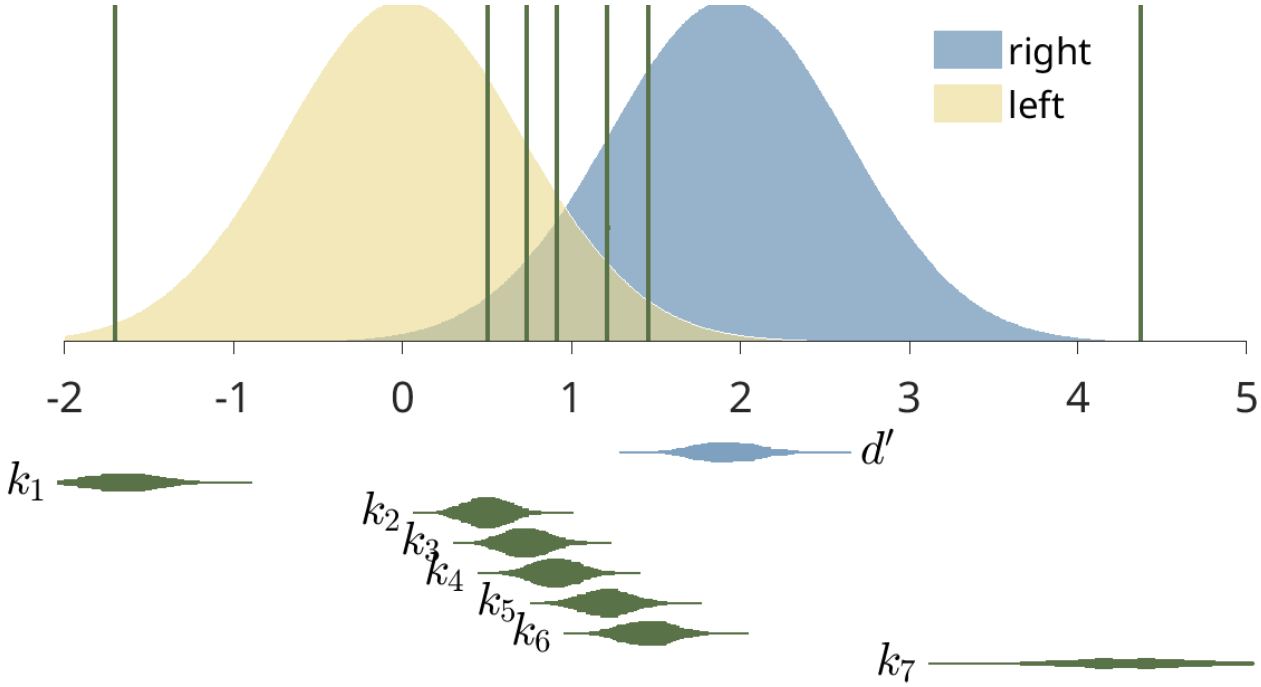
Discriminability and Criteria Inferences

- The figure shows the joint and marginal posterior distributions for discriminability d' and criteria $\boldsymbol{k}$ for the three participants



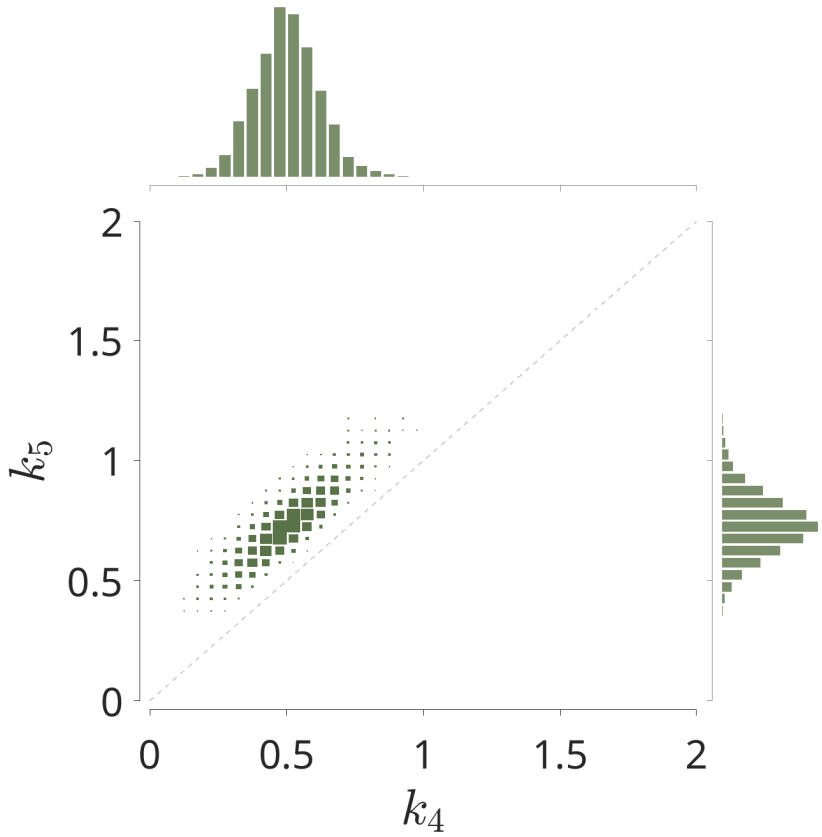
Inferences For Participant B

- The figure shows the model structure based on the posterior means and the marginal posterior distributions for discriminability d' and each of the criteria $\boldsymbol{k}$
  - explains the frequent use of confidence level 3 for both choices in terms of criteria placement



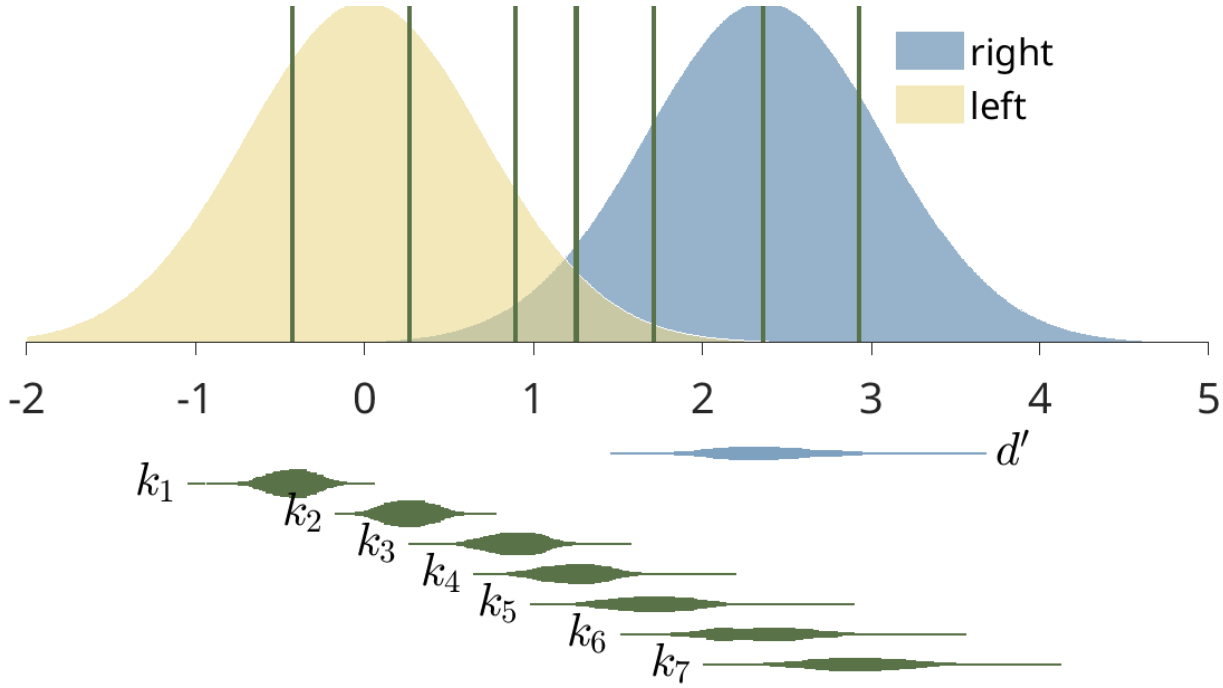
Interpreting the Criteria Inferences

- The model has the constraint that $k_1 < k_2 < k_3 < k_4$
  - the marginal posterior distributions in the interpetation visualization do not show this constraint, but it is in the joint posterior distribution
  - emphasizes that care needs to be taken in interpreting marginal posterior distributions



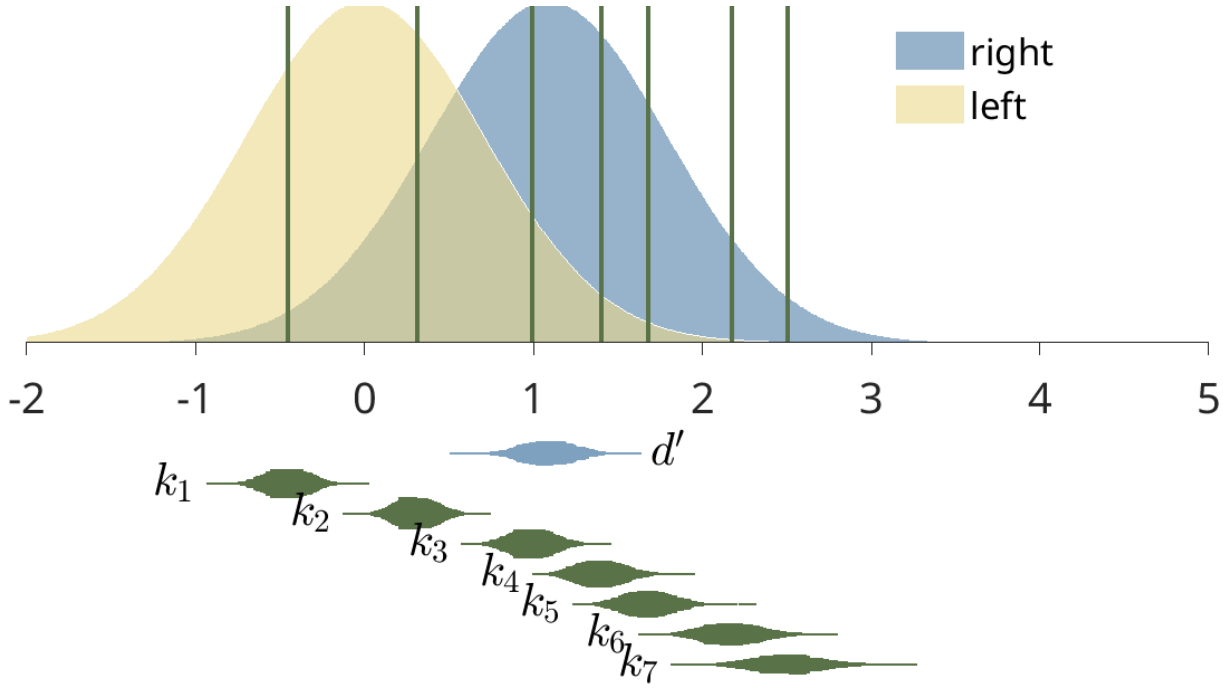
Inferences For Participant A

- The figure shows the model structure based on the posterior means and the marginal posterior distributions for discriminability d' and each of the criteria $\boldsymbol{k}$
  - shows the high discriminability and use of all confidence ratings



Inferences For Participant C

- The figure shows the model structure based on the posterior means and the marginal posterior distributions for discriminability d' and each of the criteria $\boldsymbol{k}$
  - explains the left-titled bias in terms of criteria placement and modest discriminability



References i

Shekhar, M., & Rahnev, D. (2018). Distinguishing the roles of dorsolateral and anterior pfc in visual metacognition. *Journal of Neuroscience*, *38*(22), 5078–5087. doi: 10.1101/280081